

# Sahil Tyagi

✉ Email

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🌱 ORCID

## Education

- 2018 – 2024    📖 **Ph.D., Intelligent Systems Engineering**, Indiana University Bloomington  
*Thesis: Towards Building Efficient Computation and Communication Models for Deep Learning Systems*  
*Advisor: Martin Swamy*  
*Major: Computer Engineering*  
*CGPA: 3.77/4.0*  
*Expected graduation date: August 2024*
- 2009 – 2013    📖 **Bachelor of Technology (B. Tech.), Electrical and Electronics Engineering**, Guru Gobind Singh Indraprastha University, New Delhi, India.  
*Grade: 7.6/10.0*

## Research Interests

- 📖 Large-scale ML systems
- 📖 Deep Learning
- 📖 Federated Learning
- 📖 Distributed systems
- 📖 Edge, Cloud and High-performance computing (HPC)

## Research Publications

### Journal Articles

- 1 S. Tyagi and P. Sharma, “OmniLearn: A Framework for Distributed Deep Learning over Heterogeneous Clusters,” (*under review*), 2024.
- 2 S. Chaturvedi, S. Tyagi, and Y. Simmhan, “Cost-Effective Sharing of Streaming Dataflows for IoT Applications,” *IEEE Transactions on Cloud Computing*, vol. 9, no. 4, pp. 1391–1407, 2021. 📄 DOI: 10.1109/TCC.2019.2921371.

### Conference Proceedings

- 1 S. Tyagi and M. Swamy, “Enabling Large-Batch Training via Learned Gradient Mapping,” in (*in preparation*), 2024.
- 2 S. Tyagi and M. Swamy, “On Using Large-Batches in Federated Learning,” in (*in preparation*), 2024.
- 3 S. Tyagi, “Scavenger: A Cloud Service for Optimizing Cost and Performance of DL Training,” in *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)*, Los Alamitos, CA, USA: IEEE Computer Society, May 2023, pp. 349–350. 📄 DOI: 10.1109/CCGridW59191.2023.00081.
- 4 S. Tyagi and P. Sharma, “Scavenger: A Cloud Service for Optimizing Cost and Performance of ML Training,” in *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, Accept. Rate: 21%, 2023, pp. 403–413. 📄 DOI: 10.1109/CCGrid57682.2023.00045.
- 5 S. Tyagi and M. Swamy, “Accelerating Distributed ML Training via Selective Synchronization,” in *IEEE International Conference on Cluster Computing (CLUSTER) 2023, Santa Fe, NM, USA, October 31 - Nov. 3, 2023*, Accept. Rate: 25%, IEEE, 2023, pp. 1–12. 📄 DOI: 10.1109/CLUSTER52292.2023.00008.
- 6 S. Tyagi and M. Swamy, “Accelerating Distributed ML Training via Selective Synchronization (Poster Abstract),” in *2023 IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops)*, 2023, pp. 56–57. 📄 DOI: 10.1109/CLUSTERWorkshops61457.2023.00023.
- 7 S. Tyagi and M. Swamy, “Flexible Communication for Optimal Distributed Learning over Unpredictable Networks,” in *2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy*, Accept. Rate: 17.5%, Dec. 2023.

- 8 S. Tyagi and M. Swamy, "GraVAC: Adaptive Compression for Communication-Efficient Distributed DL Training," in *16th IEEE International Conference on Cloud Computing (CLOUD) 2023, Chicago, IL, USA, July 2-8, 2023*, Accept. Rate: 20%, IEEE, 2023, pp. 319–329.  DOI: 10.1109/CLOUD60044.2023.00045.
- 9 S. Tyagi and M. Swamy, "ScaDLES: Scalable Deep Learning over Streaming Data at the Edge," in *2022 IEEE International Conference on Big Data (BigData)*, Accept. Rate: 19.2%, Los Alamitos, CA, USA: IEEE Computer Society, Dec. 2022, pp. 2113–2122.  DOI: 10.1109/BigData55660.2022.10020597.
- 10 S. Tyagi and P. Sharma, "Taming Resource Heterogeneity in Distributed ML Training with Dynamic Batching," in *2020 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS)*, Accept. Rate: 25%, Los Alamitos, CA, USA: IEEE Computer Society, Aug. 2020, pp. 188–194.  DOI: 10.1109/ACSOS49614.2020.00041.
- 11 C. Widanage, J. Li, S. Tyagi, *et al.*, "Anomaly Detection over Streaming Data: Indy500 Case Study," in *2019 IEEE 12th International Conference on Cloud Computing (CLOUD)*, Accept. Rate: 20%, 2019, pp. 9–16.  DOI: 10.1109/CLOUD.2019.00015.
- 12 J. Qiu, B. Peng, R. Teja, S. Tyagi, C. Widanage, and J. Koskey, "Real-Time Anomaly Detection from Edge to HPC-Cloud," in *2018 Big Data and Exascale Computing Workshop (BDEC2)*, 2018.  URL: [https://exascale.org/bdec/sites/exascale.org.bdec/files/whitepapers/Qiu\\_BDEC2\\_WP.pdf](https://exascale.org/bdec/sites/exascale.org.bdec/files/whitepapers/Qiu_BDEC2_WP.pdf).
- 13 S. Chaturvedi, S. Tyagi, and Y. Simmhan, "Collaborative Reuse of Streaming Dataflows in IoT Applications," in *2017 IEEE 13th International Conference on e-Science (e-Science)*, Accept. Rate: 36%, 2017, pp. 403–412.  DOI: 10.1109/eScience.2017.54.

## Skills

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|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming |  Python, C, C++, Shell scripting, MPI, OpenMP, CUDA, Java, Scala, SQL, MATLAB |
| Networking  |  Wireshark, TCPDump, Traffic Control (tc), Scapy, IPerf.                     |
| Databases   |  MySQL, MSSQL, SQLite, HBase, Neo4j                                          |
| Frameworks  |  PyTorch, TensorFlow, MXNet, Keras, Hadoop, Spark, Kafka, Slurm, Cloud APIs  |

## Teaching Experience

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|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Associate Instructor |  <b>High-Performance Computing:</b> Spring 2024                   |
|                      |  <b>Computer Networks:</b> Fall 2023, Fall 2022                   |
|                      |  <b>Operating Systems:</b> Spring 2023                            |
|                      |  <b>Engineering Distributed Systems:</b> Spring 2022, Spring 2021 |
|                      |  <b>Cloud Computing:</b> Fall 2021, Fall 2020, Fall 2019          |

## Employment History

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 – 2024 |  <b>Graduate Researcher and Associate Instructor</b> , Luddy School of Informatics, Computing and Engineering, Indiana University Bloomington, USA. |
| 2017 – 2018 |  <b>Research Staff Member</b> , Dept. of Computational and Data Sciences (CDS), Indian Institute of Science (IISc), Bengaluru, India.               |
| 2016 – 2016 |  <b>Data Scientist</b> , HT Media Limited, Gurugram, Haryana, India.                                                                                |
| 2015 – 2015 |  <b>Data Engineer</b> , Stayzilla, Bengaluru, Karnataka, India.                                                                                     |
| 2014–2015   |  <b>Software Engineer</b> , Tatra Data Limited, New Delhi, India.                                                                                   |

## Miscellaneous

### Awards and Achievements

-  **NSF Student Grant:** To present research at IEEE CLUSTER 2023, Santa Fe, New Mexico.

## Miscellaneous (continued)

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- **Luddy Dean's Graduate Student Award:** In Fall 2023 for outstanding research.
- **NSF Travel Award:** To present research at IEEE/ACM CCGrid 2023, Bengaluru, India.
- **Best early-career researcher poster award:** Awarded at IEEE/ACM CCGrid 2023.
- **Google Cloud Student Researcher (2021, 2022):** Received GCP credits for research.
- **Student Research Award:** Funded via NSF grant Data Infrastructure Building Blocks (DiBBS) 17-500, for academic year 2018-2019.

## Professional Services

- **2024: USENIX OSDI (AEC), IEEE CLUSTER (TPC), USENIX ATC (AEC)**

## Presentations and Talks

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- **4/24:** Guest lectures, "Parallel Computing with GPUs for Distributed ML Applications", High-Performance Computing (HPC) course, Indiana University Bloomington, USA.
- **12/23:** Paper presentation, "Flexible Communication for Optimal Distributed Learning over Unpredictable Networks." 2023 IEEE International Conference on Big Data, Sorrento, Italy.
- **11/23:** Paper presentation, "Accelerating Distributed ML Training via Selective Synchronization." 2023 IEEE International Conference on Cluster Computing, Santa Fe, New Mexico, USA.
- **11/23:** Poster presentation, "Accelerating Distributed ML Training via Selective Synchronization." 2023 IEEE International Conference on Cluster Computing, Santa Fe, New Mexico, USA.
- **09/23:** Invited talk, "Towards building efficient computation and communication models for distributed deep learning systems." Mathematics and Computer Science (MCS) division, Argonne National Laboratory, Illinois, USA.
- **07/23:** Paper presentation, "GraVAC: Adaptive Compression for Communication-Efficient Distributed DL Training." 2023 IEEE International Conference on Cloud Computing, Chicago, Illinois.
- **05/23:** Paper presentation, "Scavenger: A Cloud Service for Optimizing Cost and Performance of ML Training." 2023 IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, Bengaluru, India.
- **05/23:** Poster presentation, "Scavenger: A Cloud Service for Optimizing Cost and Performance of ML Training." 2023 IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, Bengaluru, India.
- **12/22:** Paper presentation, "ScaDLES: Scalable Deep Learning over Streaming Data at the Edge." 2022 IEEE International Conference on Big Data, Osaka, Japan.
- **07/20:** Paper presentation, "Taming Resource Heterogeneity in Distributed ML Training with Dynamic Batching." 2020 IEEE International Conference on Autonomic Computing and Self-Organizing Systems, virtual.
- **11/18:** "Real-Time Anomaly Detection from Edge to HPC-Cloud", Intel Speakerships at SC18 (Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis 2018), Dallas, Texas, USA.

## References

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Available upon request